

Anatomically-Constrained LDMs for Rare Case Augmentation

Project 3

Computer Vision - Prof. Amerini

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Table of Contents

1 Problem Statement

► Problem Statement

► State of the Art

► Proposed Method

► Dataset

► Experimental Setup

► Evaluation

► Results

► Conclusions

► References



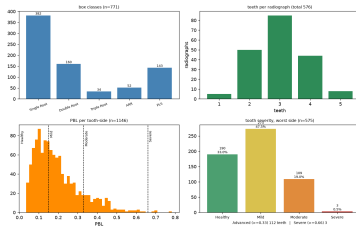
Problem Statement

1 Problem Statement

Periodontitis grading from a periapical X-ray is heavily imbalanced: severe bone loss is rare in every public dataset, including ours: **3 Severe teeth out of 575**.

Standard augmentation (rotation, flip) cannot create new **anatomical variations**; models overfit to healthy/mild cases and fail on advanced disease.

Mask-based augmentation. Bone loss is a *geometric* property. Conditioned on an anatomical mask, a severe case can be commanded by moving the crest line down, even without ever having seen one.



Research questions:

1. Does latent diffusion degrade anatomical precision?
2. Does an anatomical loss recover it?
3. Does the synthetic data help a downstream classifier?



Table of Contents

2 State of the Art

► Problem Statement

► State of the Art

► Proposed Method

► Dataset

► Experimental Setup

► Evaluation

► Results

► Conclusions

► References



State of the Art

2 State of the Art

DDPMs (Ho et al., 2020) and LDMs (Rombach et al., 2022) are standard for medical image generation, but often hallucinate fine anatomical elements (Kazerouni et al., 2023).

Konz et al., MICCAI 2024: segmentation-guided diffusion, with mask concatenated at every denoising step, mask-ablated training. Dice 0.90 on breast MRI/CT, ~12000 training slices, **in pixel space**.

Their latent baseline (fine-tuned Stable Diffusion + ControlNet) **fails**: Dice 0.36 and 0.11. They warn that a latent space risks losing *"precise spatial guidance"*.

Open question. *"we did not compare to ControlNet-like latent diffusion models trained completely from scratch."*



Table of Contents

3 Proposed Method

- ▶ Problem Statement
- ▶ State of the Art
- ▶ **Proposed Method**
- ▶ Dataset
- ▶ Experimental Setup
- ▶ Evaluation
- ▶ Results
- ▶ Conclusions
- ▶ References



Proposed Method: the anatomical mask

3 Proposed Method

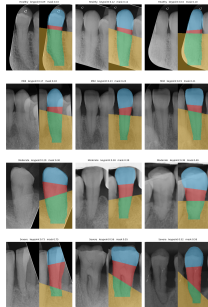
Each tooth is rotated to vertical, canonicalised crown-up, cropped to 1:2 with margin, resized to 256×128 . A 5-class map is built from the keypoints:

-
- 0 background / soft tissue
 - 1 crown
 - 2 **exposed root** ← the signature
 - 3 embedded root
 - 4 alveolar bone
-

Severity becomes geometry:

$$\text{PBL} = \frac{\gamma_{\text{crest}} - \gamma_{\text{CEJ}}}{\gamma_{\text{apex}} - \gamma_{\text{CEJ}}}$$

Moving the crest line down commands a severe case.



Tooth silhouette from S_build (binary UNet, val Dice 0.9221, trained on 8977 auxiliary patches).



Proposed Method — models and the anatomical loss

3 Proposed Method

Two-stage LDM. Frozen AutoencoderKL ($f=4, 256 \times 128 \rightarrow 64 \times 32 \times 4$) + UNet2DModel denoiser, both randomly initialised. Mask one-hot to 5 channels, concatenated at every step. Mask-ablated training with $p_{\text{class}}=0.25$, $p_{\text{null}}=0.1$. Identical DDPM baseline in pixel space, same capacity ($\sim 29\text{M}$ params), same schedule.

Original contribution — an explicit anatomical loss. Decode the x_0 -prediction, push it through a frozen segmenter, penalise disagreement with the *commanded* mask:

$$\mathcal{L} = \text{MSE}(\hat{c}, \epsilon) + \lambda \cdot \mathcal{L}_{\text{seg}}(\text{S_eval}(\mathcal{D}(\hat{x}_0)), m)$$

Applied only at low t , only where the mask survived ablation. λ from gradient-norm matching (Esser et al. 2021) — not hand-tuned.

Independent evaluation. S_eval is also the project's evaluator. So a **third** segmenter S_eval2 is trained, never inside any loss, and every result is reported under both.



Table of Contents

4 Dataset

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ **Dataset**
- ▶ Experimental Setup
- ▶ Evaluation
- ▶ Results
- ▶ Conclusions
- ▶ References



Dataset — perio-KPT

4 Dataset

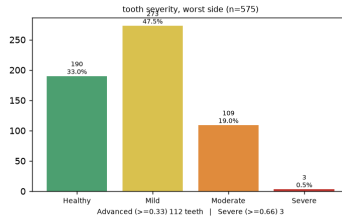
192 periapical radiographs, 771 annotated teeth, 11 keypoints each (CEJ, bone level, root apex per side).

Correction to the brief: the 5 label classes are tooth *morphologies*, not periodontitis stages. Severity had to be **computed** from the keypoints.

575 usable teeth, severity = worst side:

Healthy	190	33.0%
Mild	273	47.5%
Moderate	109	19.0%
Severe	3	0.5%

Reproduced the clinicians' own count: 69 affected images vs. their 70.



Downstream rare class: **Advanced** = $PBL \geq 0.33$, 112 teeth (19.5%). *Severe alone is not trainable.*

S_eval's own floor on real images: PBL MAE **0.0436**, Dice **0.8127**.



Table of Contents

5 Experimental Setup

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ Dataset
- ▶ **Experimental Setup**
- ▶ Evaluation
- ▶ Results
- ▶ Conclusions
- ▶ References



Experimental Setup

5 Experimental Setup

Model	Role	Params
S_build	tooth silhouette	7.8M
S_eval	faithfulness judge	7.8M
S_eval2	second judge	7.8M
VAE	$f=4$ compression	3.5M
LDM	the deliverable	29.1M
DDPM-px	pixel baseline	29.3M
ResNet-18	downstream classifier	11.2M

Splits are image-level, stratified by severity — patches from one radiograph share exposure and anatomy, so patch-level splitting would leak.

134 / 19 / 39 images $\rightarrow$ 395 / 54 / 126 patches.

Five hard gates, each minutes, each preventing hours:

1 PBL reproduction	PASS
2a crest not singled out	PASS
2b VAE not blurred flat	FAIL
3 mask $\leftrightarrow$ keypoints	PASS
4 overfit one batch	PASS
5 mask reaches denoiser	PASS

Gate 2b (0.706 vs. 0.75) was **not** chased: the VAE round-trip was measured to cost only 0.008 Dice, so the bottleneck was elsewhere.

All of it on one **8 GB** laptop GPU.



Table of Contents

6 Evaluation

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ Dataset
- ▶ Experimental Setup
- ▶ **Evaluation**
- ▶ Results
- ▶ Conclusions
- ▶ References



Evaluation protocol

6 Evaluation

Metric interpretation. Interpret generated-image scores against real-image and null-condition baselines.

Measurement floor S_{eval} scores PBL MAE **0.0436** on *real* images. Nothing below that is interpretable.

Ceiling Real images score Dice **0.8127** on exposed root through the same network. *Konz reports 0.9027 with no such denominator.*

Chance anchor The null condition still scores Dice 0.293 — without it, 0.29 would look like partial success.

Two judges S_{eval} and S_{eval2} agree on real images (0.0382 vs. 0.0428) and diverge on generated ones (0.1377 vs. 0.2012). **Model differences are robust; absolute levels are not.**

Paired tests Seeds and perturbations are matched, so systematic error cancels.



Table of Contents

7 Results

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ Dataset
- ▶ Experimental Setup
- ▶ Evaluation
- ▶ **Results**
- ▶ Conclusions
- ▶ References



Results: latent and pixel diffusion

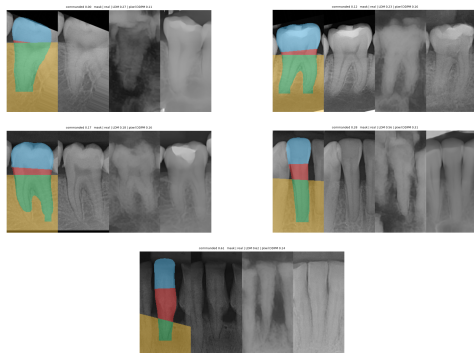
7 Results

	LDM latent	DDPM pixel	real ceiling
Dice, exposed root	0.5915	0.7572	0.8127
Dice, mean	0.8069	0.8849	0.9079
PBL MAE	0.1377	0.0555	0.0382
PBL correlation	+0.382	+0.698	+0.876
PBL bias	+0.121	+0.003	+0.000
FID	249.9	139.8	—

Decomposition:

real 0.8127 $\rightarrow$ VAE round-trip 0.8043 $\rightarrow$ latent diffusion 0.5915.

The compression is nearly free. The loss is the **diffusion in the latent space**.



Within each panel: mask, real, LDM, DDPM.
All five examples, in reading order.



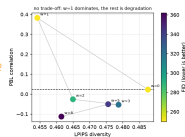
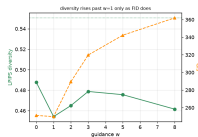
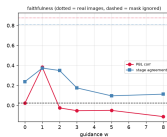
Ablation: classifier-free guidance

7 Results

w	Dice cl.2	PBL MAE	LPIPS	FID
0.0	0.2927	0.2502	0.488	251.2
1.0	0.5915	0.1377	0.454	249.9
2.0	0.4125	0.2150	0.465	289.5
5.0	0.1479	0.5032	0.476	342.3
8.0	0.1520	0.5218	0.461	361.9

Best performance at $w = 1$. Higher guidance increases PBL error and FID; exposed-root Dice remains below the $w = 1$ result.

LPIPS varies little (0.454–0.488; real patches: 0.551). No clear fidelity/diversity trade-off is observed.





Results: anatomical loss

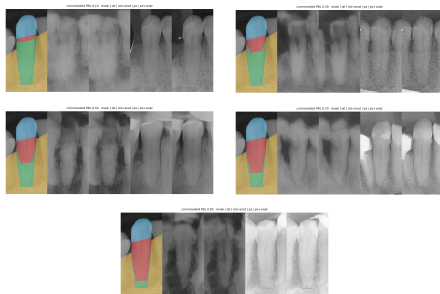
7 Results

Apply the same anatomical loss to both spaces for a matched comparison.

Dice, exposed root	latent	pixel	gap
plain mask conditioning	0.5915	0.7572	0.1657
+ anatomical loss	0.7358	0.9001	0.1643

Comparison. Both spaces gain approximately 0.14 Dice. The latent-pixel gap remains near 0.16.

The independent judge (S_eval2) confirms the gain. Pixel FID is essentially unchanged: 139.8 $\rightarrow$ 138.0.



Within each panel: mask, latent, latent + anatomy, pixel, pixel + anatomy. All five commands, in reading order.



Severity control and downstream classification

7 Results

Counterfactual. 16 teeth, crest commanded to PBL 0.10–0.85, i.e. masks no model has seen.

model (judge S _{eval2})	slope	monotone
latent, plain	0.137	3/16
latent + anat.	0.380	3/16
pixel, plain	0.211	0/16
pixel + anat.	0.259	8/16

Severity control remains incomplete. The best slope is 0.38; no model reliably follows the requested severity.

Downstream (H2). 11 arms $\times$ 5 seeds, paired per-seed differences in mAP:

synthetic vs. real-only	$+0.037 \pm 0.041$
guidance vs. no guidance	$+0.013 \pm 0.032$

Small average gain from synthetic data; no detectable benefit from guidance.

Only **24 Advanced teeth** in the test set, limiting the ability to resolve small performance differences.



Table of Contents

8 Conclusions

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ Dataset
- ▶ Experimental Setup
- ▶ Evaluation
- ▶ Results
- ▶ **Conclusions**
- ▶ References



Conclusions

8 Conclusions

1. **Latent-pixel gap.** A from-scratch latent LDM reaches Dice 0.807 mean (his latent baseline: 0.11–0.36), but stays 0.16 Dice behind an identical pixel model on the exposed root — and an explicit constraint given to *both* does not close that gap.
2. **Anatomical loss improves both models.** +0.14 Dice, FID unchanged, and it regularises: unconstrained runs overfit from step 5000, the constrained ones do not.
3. **Data limitations.** 3 Severe teeth in 575. No model commands a severity it has not seen; the downstream task cannot resolve the difference with 24 Advanced test teeth.

Synthetic labels. Labelling synthetic images by observed rather than requested severity improved macro-F1 in all four banks.

Future work. Anatomy-supervised autoencoder; SPADE-style mask injection instead of channel concatenation; a test set with enough rare cases to make the downstream question answerable at all.



Table of Contents

9 References

- ▶ Problem Statement
- ▶ State of the Art
- ▶ Proposed Method
- ▶ Dataset
- ▶ Experimental Setup
- ▶ Evaluation
- ▶ Results
- ▶ Conclusions
- ▶ **References**



References

9 References

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Thank you for listening!